

Student Spending & Expense Behavior Report

1. Problem Statement

The objective of this project is to analyze how students allocate their monthly expenses and understand their financial behavior patterns. The study focuses on identifying how spending varies across different categories, demographic groups, and behavioral factors such as impulse purchasing and expense tracking.

This analysis is intended for:

- **Students**, to improve financial decision-making
- **Educational institutions**, to design effective financial awareness initiatives

2. Objective of the Project

The key objectives of this analysis are:

- To understand how students distribute their expenses across different categories
- To examine the relationship between income and total spending
- To analyze how behavioral factors such as impulse purchasing and expense tracking influence financial outcomes
- To evaluate the impact of financial awareness on savings
- To identify patterns in spending behavior over time

3. Data Description

The dataset consists of **131 student responses** collected through a structured questionnaire. It integrates financial, demographic, and behavioral variables to provide a comprehensive understanding of student financial habits.

Types of Variables

Numerical Variables:

- Income
- Age
- Total_Spending
- Savings
- Financial_Awareness

Categorical Variables:

- Gender
- Living_Arrangement
- Expense_Tracking
- Impulse_Purchase
- End_Month_Behavior
- Income_Sufficiency
- Financial_Status

Spending Categories:

- Food
- Transport
- Shopping
- Entertainment
- Academic

Time-Based Variables:

- Expense_Date
- Expense_Time

Derived Columns

- **Total_Spending** = Sum of all expense categories
- **Savings** = Income – Total_Spending
- **Financial_Status** = Saver / Break Even / Overspender

- **Financial_Perception_Accuracy** = Match between perceived and actual financial behavior

This structured dataset supports **multi-dimensional analysis** across financial, behavioral, and demographic aspects.

4. Data Collection Method

The data was collected through a **primary survey questionnaire** distributed among students.

- **Mode:** Online form
- **Type:** Self-reported responses
- **Nature:** Cross-sectional data
- **Sample Size:** 131 respondents

The questionnaire included demographic details, category-wise expenses, behavioral indicators, and financial awareness measures.

5. Power Query Steps

Data preprocessing was performed to ensure accuracy and usability:

- Removed unnecessary columns (e.g., email)
- Standardized column names
- Cleaned text fields (trimmed spaces, removed hidden characters)
- Corrected data types
- Created calculated columns:
 - Total_Spending
 - Savings
 - Financial_Status
 - Financial_Perception_Accuracy

Category Table Transformation

To enable category-wise analysis:

- Duplicated the main table
- Retained only category-related columns
- Unpivoted data into:
 - Category
 - Amount
- Created relationship using **Response_Timestamp**

This ensured proper interaction between slicers and visuals.

6. DAX Measures Used

- **Total Students** – Counts the total number of student records in the dataset
- **Average Income** – Calculates the mean monthly income of students
- **Average Spending** – Calculates the mean total spending
- **Average Savings** – Calculates the mean savings
- **Savings Rate** – Measures savings as a proportion of income
- **Spending Ratio** – Measures spending relative to income
- **Savers % / Overspenders %** – Represents the percentage distribution of financial behavior categories
- **Accuracy %** – Measures the percentage of students whose financial perception matches their actual financial behavior
- **Average Awareness** – Calculates the mean financial awareness score
- **Dynamic Titles & Subtitles** – Implemented using DAX to reflect slicer selections (such as Gender, Living Arrangement, Expense Tracking, and Impulse Purchase), enhancing interactivity and contextual understanding of visuals

7. Dashboard Explanation

Page 1: Student Spending & Financial Overview

1. Total Spending by Category (₹)

- **Question:** How do students allocate their expenses?
- **Type:** Comparison
- **Fields:** Category, Amount (SUM)

- **What it shows:** Food has the highest total spending, followed by shopping and transport
- **Insight:** Essential expenses dominate overall student spending

2. Distribution of Financial Status (%)

- **Question:** What is the financial condition of students?
- **Type:** Distribution
- **Fields:** Financial_Status, Total_Students
- **What it shows:** Majority of students are Savers (~72%), with fewer Overspenders (~14%)
- **Insight:** Most students maintain a positive financial balance

3. Average Spending by Gender (₹)

- **Question:** Does spending differ by gender?
- **Type:** Comparison
- **Fields:** Gender, Total_Spending (AVERAGE)
- **What it shows:** Male students have slightly higher average spending than female students
- **Insight:** Gender-based differences exist but are not highly significant

4. Average Spending by Living Arrangement (₹)

- **Question:** How does living arrangement affect spending?
- **Type:** Comparison
- **Fields:** Living_Arrangement, Total_Spending (AVERAGE)
- **What it shows:** PG students spend the most, while hostel students spend the least
- **Insight:** Greater independence leads to higher spending

5. Income vs Total Spending (₹)

- **Question:** What is the relationship between income and spending?
- **Type:** Relationship
- **Fields:** Income (X-axis), Total_Spending (Y-axis), Financial_Status (Legend), Savings (Size), Response_Timestamp (Value)
- **Additional Context (Tooltip):** Living_Arrangement, Impulse_Purchase, Expense_Tracking
- **What it shows:** A clear positive correlation between income and total spending, with variation across financial behavior groups
- **Insight:** Higher income leads to proportionally higher spending rather than increased savings

6. Average Spending by Impulse Purchase Frequency (₹)

- **Question:** How does impulse buying affect spending?
- **Type:** Comparison
- **Fields:** Impulse_Purchase, Total_Spending (AVERAGE)
- **What it shows:** “Often” impulse buyers have the highest average spending
- **Insight:** Impulse behavior is a key driver of increased spending

7. Average Savings by Expense Tracking Habit (₹)

- **Question:** Does expense tracking improve savings?
- **Type:** Comparison
- **Fields:** Expense_Tracking, Savings (AVERAGE)
- **What it shows:** Students who do not track expenses show slightly higher savings
- **Insight:** Expense tracking alone does not ensure higher savings, indicating that behavioral discipline and spending control play a more critical role

8. Impact of Financial Awareness on Savings (₹)

- **Question:** Does financial awareness influence savings?
- **Type:** Trend
- **Fields:** Financial_Awareness, Savings (AVERAGE)
- **What it shows:** Savings increase consistently with higher awareness levels
- **Insight:** Financial awareness shows a strong positive relationship with savings, indicating better financial decision-making

9. Spending Trend Over Time

- **Question:** How does spending vary over time?
- **Type:** Trend
- **Fields:** Expense_Date, Total_Spending (AVERAGE), Savings (AVERAGE)
- **What it shows:** Spending and savings fluctuate with noticeable peaks on certain days
- **Insight:** Spending shows distinct peaks on certain dates, indicating that student expenses are event-driven rather than consistently distributed over time

Tooltip: Average Spending by Time of Day (₹)

- **Question:** When do students spend the most?
- **Type:** Comparison
- **Fields:** Expense_Time, Total_Spending (AVERAGE)
- **What it shows:** Spending varies across Morning, Afternoon, Evening, and Night depending on context
- **Insight:** Spending patterns are situational rather than fixed

8. Key Insights

1. Food accounts for the highest share of total spending, indicating that essential expenses dominate student budgets.
2. A strong positive relationship exists between income and spending, suggesting that higher income does not necessarily lead to higher savings.
3. Impulse purchasing significantly increases spending, with frequent impulse buyers showing the highest average expenditure.
4. Financial awareness shows a consistent positive relationship with savings, indicating that informed students manage finances more effectively.
5. Living arrangement influences spending behavior, with students living independently (PG) spending more than those in hostels.
6. Expense tracking alone does not ensure higher savings, highlighting the importance of discipline and financial planning.

9. Recommendations

The following recommendations are directly derived from the insights obtained through the analysis:

1. **Enhance financial awareness programs**
→ Conduct workshops on budgeting and financial literacy to improve saving behavior
2. **Encourage controlled spending habits**
→ Students should limit impulse purchases by setting spending limits and using budgeting tools
3. **Promote structured financial planning**
→ Students should move beyond tracking and actively plan their expenses for better financial outcomes

10. Conclusion

This analysis demonstrates that student financial behavior is influenced by a combination of income, lifestyle, and behavioral factors. While the majority of students are able to save, risks such as impulse spending and low financial awareness can negatively affect financial stability.

Improving financial awareness and encouraging disciplined spending habits can significantly enhance students' financial well-being and long-term financial management.